ADITYA KUMAR

Machine Learning Engineer | Production AI & GenAI Systems

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 2+ years of experience designing, training, evaluating, and deploying end-to-end ML and Generative AI systems in production environments. Proven track record building time-series forecasting pipelines, automating model benchmarking and selection with MAPE-based evaluation, and shipping serverless ML workloads on AWS Lambda. Hands-on experience with LLMs, Retrieval-Augmented Generation (RAG), and agentic AI, plus computer vision and IoT-based systems. Delivered measurable business impact: 40+ hours/week saved through pipeline automation, 20% improvement in targeting efficiency, and 45% reduction in data latency.

CORE TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++
Machine Learning: Scikit-learn, TensorFlow, Time-Series Forecasting, Feature Engineering, Model Evaluation & Selection, Statistical Modeling, NLP, Computer Vision (YOLOv8)
Generative AI: LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering & Orchestration, Multi-Step Reasoning Agents, Memory-Based Context
MLOps & Cloud: CI/CD Pipeline Automation, Model Deployment & Monitoring, AWS Lambda, AWS EC2, Docker, Oracle Cloud Infrastructure (OCI), IoT Core
Data & Modeling: PostgreSQL, Oracle DB, ETL, Data Validation & Preprocessing, MAPE, Dirichlet-Multinomial (DirMult) Modeling, Hierarchical Clustering

PROFESSIONAL EXPERIENCE

Genpact — Data Scientist, Bengaluru, India Aug 2024 – Present
End-to-End ML Pipelines: Architected Python ML pipelines to ingest, clean, and preprocess enterprise time-series data; engineered predictive features and trained statistical/ML models for production forecasting workloads deployed on AWS Lambda.
Model Evaluation & Selection: Benchmarked competing forecasting models using MAPE and automated best-model selection logic, integrating results into CI/CD pipelines for repeatable, auditable ML deployment cycles.
Enterprise Modeling Platform: Built PostgreSQL-based analytical workflows combining client-specific attribute/value filters, time-period and geography segmentation, and recoding logic with Dirichlet-Multinomial (DirMult) modeling and hierarchical clustering to generate CDT decision trees and downstream analytical outputs.
Automation Impact: Automated preprocessing, validation, and pre-QC workflows in Python and R, eliminating ~40 hours/week of manual effort and improving data pipeline reliability.
Applied ML for Business: Developed Scikit-learn customer segmentation and behavioral prediction models that improved targeting efficiency by 20%, directly informing marketing and business strategy.
GenAI Prototyping: Built a goal-driven LLM analytics agent using agentic AI and RAG patterns, enabling multi-step reasoning, tool invocation, and automated insight generation over enterprise datasets.
Genpact — Data Analyst Intern, Bengaluru, India Jan 2024 – Jun 2024

- Optimized complex SQL queries against Oracle Database, reducing dashboard data-fetch latency by 45% and improving reporting responsiveness for business stakeholders.

Designed and delivered interactive KPI dashboards in Power BI and Tableau to support data-driven business reporting.

SELECTED PROJECTS

LLM-Based Analytics Agent (Internal POC): Built a Python agentic AI system with multi-step reasoning, tool usage, prompt orchestration, and memory-based context for automated data analysis and insight generation using RAG patterns.
RoboWeed – IoT Agricultural Rover: Engineered a real-time YOLOv8 computer-vision pipeline on Raspberry Pi with Python, Roboflow, and IoT sensors for automated weed detection and targeted spraying; achieved 90% detection accuracy.
Sales-Nexus – Time-Series + LLM Forecasting Framework: Designed a reusable forecasting framework combining preprocessing, feature engineering, error-based model selection, and LLM-generated business insights.

EDUCATION

RV Institute of Technology and Management — B.E., Computer Science 2020 – 2024
GPA: 8.04 / 10

CERTIFICATIONS

Lean Six Sigma Green Belt — Genpact
Machine Learning, Data Science and Generative AI with Python — Udemy
Natural Language Processing with Python — Udemy
Oracle Cloud Infrastructure (OCI) AI Foundations — Oracle University
Python Data Structures — University of Michigan (Coursera)

ACHIEVEMENTS

LeetCode Rating: 1835 (Top 6.3% globally); Qualified for Google Code Jam (Rank 71/100 in round)
Co-Lead, Competitive Programming — Google Developer Student Clubs (DSC), RVITM
1st Place, CODE BATTLE and 2nd Place, CODM — Phoenix Robotics Club